Conditional diffusion for seismic waveforms: model design and evaluation at ARMT

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Abstract

Generating seismic waveforms requires a model to reproduce both event-dependent patterns and variation among recordings. We study whether an event-dependent reference distribution improves waveform diffusion at a single station, and whether the conclusion depends on how the generated waveforms are evaluated. The ARMT archive contains 1,183 vertical recordings; 774 are used for training and 169 for evaluation. Across three runs, conditioning the reference on event characteristics reduces the mean feature-based continuous ranked probability score from 0.4933 to 0.4618, a 6.39% improvement over a global reference. The difference remains after filtering generated waveforms. An empirical nearest-neighbor model obtains a better feature score, but conditional diffusion obtains better waveform energy, variogram and time-frequency scores. Filtering further improves the neural models' feature scores while slightly worsening some waveform scores. These results show that a simple conditional reference can help waveform diffusion, but good feature scores do not guarantee good waveform scores. The evaluation set was also used for model selection, so the findings are exploratory and require confirmation on independent recordings.

Keywords: seismic waveforms; conditional diffusion; stochastic processes; probabilistic evaluation; scoring rules.

1 Introduction

A seismic waveform records how ground motion changes over time. Its amplitude, arrival pattern, duration and frequency content depend on the earthquake and its location relative to the recording station. Even events with similar recorded characteristics can produce different waveforms. A useful generator must therefore describe a distribution of plausible recordings, rather than predict a single trace.

This problem is difficult when only a small archive is available at one station. A flexible model must learn both broad event-dependent structure and the remaining waveform variation. It is also difficult to evaluate. Two generators may reproduce similar amplitude distributions while differing in timing or temporal dependence. A ranking based on summary features can therefore differ from a ranking based on the full waveform.

Recent seismic generators use diffusion and related probabilistic models to describe these distributions. HighFEM generates latent spectrogram representations with a diffusion model (Palgunadi et al., 2025). SeisCFM uses conditional flow matching to generate spectrograms and reconstruct three-component ground motions (Gu et al., 2026). These studies establish useful approaches, but their recording regions, components and processing choices differ from those of a single-station archive. Their published scores are therefore not directly comparable with scores calculated here.

We examine a narrower question: does an event-dependent reference help a fixed waveform diffusion model? The reference predicts amplitude, the time envelope (the rise and decay of amplitude), and spectral shape for an event. An invertible transformation uses these predictions to rescale the waveform before diffusion training. Generation reverses the transformation. The comparison model uses a global reference with the same transformation structure. Both networks receive the same event characteristics, so the comparison isolates the choice of reference within an otherwise shared modeling procedure.

This design builds on established work. Karras et al. (2022) develop a common framework for diffusion training, preconditioning and sampling. Lee et al. (2022) show how a conditional, data-dependent Gaussian prior can improve speech generation. Here, we use a conditional envelope and spectrum to define waveform coordinates and evaluate their practical effect on the ARMT archive. Our contribution is to measure how this reference affects generation at one station and how the result depends on the evaluation measure.

Conditional reference diffusion improves on global reference diffusion across three runs. However, an empirical nearest-neighbor model achieves a better score on the selected waveform features, while diffusion performs better on three complementary waveform scores. A filtering comparison changes the feature ranking again. Together, these findings show why model design and distributional evaluation must be considered jointly.

The paper proceeds from data and model design to evaluation, results and interpretation. A supplementary comparison with a latent-spectrogram adaptation of HighFEM appears in Appendix E.

2 Data and study design

2.1 The ARMT archive

The archive contains 1,183 vertical-component recordings from station KO.ARMT. The events span 2013–2024 and have reported magnitudes from 1.5 to 3.9. Each original trace contains 7,001 samples at a sampling rate of 100 Hz. We retain the first 7,000 samples, remove each trace's mean and downsample with an antialiasing filter. The resulting series contains 3,500 samples at 50 Hz and represents a 70-second window.

Each recording has seven conditioning variables: magnitude, log-transformed reported distance, log-transformed depth, latitude, longitude, and the sine and cosine of azimuth. The distance and depth transformations are $\log(1 + d)$. Distance is used as reported in the archive; its geometric convention is not independently established. All estimated waveform scales, metadata scales and model parameters use training records only.

The archive's processing history is incomplete. Comparisons with three corresponding public recordings support a window beginning ten seconds before the event origin and amplitudes consistent with filtered velocity. Those matches do not establish physical calibration or the exact filter for every recording. We therefore report amplitude-dependent scores in archive units. The analysis concerns the statistical properties of the available traces, not calibrated acceleration or structural response.

2.2 Temporal partition and evaluation scope

The training period ends before 1 July 2020. The validation period starts on that date and ends before 1 January 2023. Later recordings form a separate temporal partition. To reduce overlap

among nearby events, we connect events separated by at most seven days and ten kilometers and take the connected components as proximity groups. Earlier records are excluded when their group also appears in a later partition. This gives the counts in Table 1.

Table 1: Archive partition. Proximity groups are connected components under the seven-day and ten-kilometer rule. Only the training and validation partitions enter the results reported here.

Partition	Recordings	Groups	Role in this study
Training	774	420	Parameter estimation
Validation	169	99	Model selection and evaluation
Later temporal partition	237	99	Not evaluated here
Boundary exclusions	3	—	Excluded

The validation set serves two purposes. A subset is used to choose model checkpoints, and the full set is used to compare generated distributions. It has also informed model development. The reported performance is therefore exploratory. It is not an independent test estimate. The later temporal partition is excluded from this study because it was used in earlier model assessment.

The grouping is a practical way to account for some event dependence. It is not a validated classification of earthquake families, and dependence may remain between groups. These qualifications apply to both the mean comparisons and their uncertainty intervals.

3 Conditional waveform models

3.1 An event-dependent reference

Let $x \in \mathbb{R}^T$ denote a waveform after a common training-based amplitude normalization, and let c denote its event characteristics. We describe broad waveform structure with two positive functions: $E(c)$ gives amplitude over time, and $S(c)$ gives spectral amplitude. Together they define the linear operator

$$L(c) = D_{E(c)} F^* D_{S(c)} F, \qquad z = L(c)^{-1} x, \tag{1}$$

where D denotes a diagonal operator and F is the orthonormal Fourier transform. The spectral function is symmetric so that the transformation maps real waveforms to real waveforms. In computation, a real FFT and its inverse implement the same operation.

Equation (1) has a simple interpretation. To obtain z, first divide the waveform by its predicted time envelope, then divide its Fourier coefficients by the predicted spectral shape. Generation reverses these steps. Positivity makes the transformation invertible, so this change of coordinates does not discard waveform information. The transformed data may still have temporal dependence and a non-Gaussian distribution.

The envelope includes a predicted overall amplitude and a normalized time-varying shape. We estimate log amplitude, 64 envelope summaries and 64 spectral summaries using ridge regression on event characteristics. The regressors include linear and quadratic terms. Predictions are shrunk toward their training means, and the shape ranges are bounded to avoid extreme rescaling. Spectral targets are calculated after dividing training waveforms by their predicted envelopes. Appendix A gives the numerical settings.

The *conditional reference* uses these event-dependent predictions. The *global reference* uses intercept-only predictions in the same reference family. In both cases, diffusion learns the remaining distribution in the transformed coordinates. If z were standard Gaussian, the induced

waveform covariance would be $L(c)L(c)^{\mathsf{T}}$. The trained diffusion model can represent departures from that Gaussian law.

At generation time, the reference depends only on c and training estimates. It does not use the target recording, its observed amplitude or any partial waveform. The model must therefore learn amplitude variability from the training archive.

3.2 Diffusion in the transformed coordinates

For each reference, we divide z by a training-based scalar a equal to twice its pooled root mean square. This gives training coordinates $u = z/a$ with pooled RMS 0.5. We use the Elucidated Diffusion Model (EDM) framework of Karras et al. (2022). Its denoiser D_θ is trained using

$$\mathcal{L}(\theta) = \mathbb{E}\left[w(\sigma)\frac{1}{T}\left\|D_\theta(u + \sigma\epsilon, \sigma, c) - u\right\|_2^2\right], \qquad \epsilon \sim N(0, I), \tag{2}$$

where the noise distribution and weight w are specified in Appendix A. Samples from the learned distribution are multiplied by a, transformed back through $L(c)$, and returned to archive amplitude units.

Both reference models use the same multiscale convolutional network (a one-dimensional U-Net), event inputs, optimization settings and sampling method. The network has base width 64, residual blocks, attention at its coarsest resolution, and explicit time-position channels. Training uses batches of 16, an initial learning rate of 10^{-4}, cosine learning-rate decay and exponential moving averages of model weights. Each run contains 10,000 training steps.

Every 2,000 steps, eight waveforms are generated for each of 64 evenly spaced records from the ordered validation set. The checkpoint with the lowest primary feature score is retained. Final evaluation uses 16 draws for each of the 169 validation events and 25 sampling steps. All three conditional runs select step 8,000; all three global runs select step 10,000. The comparison therefore includes the same checkpoint-selection rule rather than imposing the same selected training step.

There are three runs of each neural model. Training and generation randomness change together. The first pair uses PyTorch 2.10, whereas the second and third pairs use PyTorch 2.11; all use Tesla T4 GPUs. Each conditional/global pair uses the same software version. Differences across pairs may reflect both software versions and random seeds.

3.3 Empirical nearest-neighbor generation

A simple empirical model provides a useful comparison because it retains the observed shape of actual training waveforms. For a query c, we find the 32 nearest training records in standardized metadata distance. If d_j is a selected distance, its sampling weight is

$$p_j(c) = \frac{\exp[-d_j^2/(2h^2)]}{\sum_{\ell=1}^{32}\exp[-d_\ell^2/(2h^2)]}, \qquad h^2 = \max\left\{\max_j d_j^2, 10^{-8}\right\}. \tag{3}$$

We draw complete training waveforms with replacement from this distribution. The model does not rescale a selected waveform using the target event's observed amplitude.

The empirical model generates a conditional distribution supported on the training archive. It can reproduce detailed recorded shapes, but cannot create a new shape outside that support. We evaluate three sets of 16 draws per event. These repetitions vary sampling only; they are not independent neural fits. The first uses GPU sampling and the other two use CPU sampling under the same probability law.

4 Evaluating generated distributions

4.1 Feature-based CRPS

An individual generated waveform need not match the exact phase of the observed trace. We therefore evaluate distributions of generated outcomes rather than the error of a single reconstructed recording. The main measure is the continuous ranked probability score (CRPS), applied to interpretable waveform features.

For an observed scalar feature y and K independently generated values $v_1, \ldots, v_K$, the fair ensemble estimate is

$$\widehat{\mathrm{CRPS}} = \frac{1}{K}\sum_{k=1}^{K}|v_k - y| - \frac{1}{2K(K-1)}\sum_{k\neq\ell}|v_k - v_\ell|. \tag{4}$$

The first term measures distance to the observation; the second accounts for spread among the generated values. The $K(K-1)$ correction targets the population score under independent ensemble sampling (Ferro, 2014). Lower scores are better.

Features are standardized using their training means and standard deviations. We first average feature scores within each of five groups, then give each group equal weight. The groups measure amplitude, energy-based timing, spectral content, the time envelope and autocorrelation. Appendix B defines them precisely. Equal group weighting prevents the larger groups from dominating solely because they contain more features.

This score is informative but incomplete. It evaluates the marginal distributions of selected features. It does not identify every aspect of the joint waveform distribution. A generator can therefore improve this score while performing worse on another valid measure.

4.2 Complementary waveform scores

We use three additional measures to assess the distribution from different perspectives. The *waveform energy score* replaces the absolute scalar distances in Equation (4) with Euclidean distances between full waveforms and divides by $\sqrt{T}$. It measures the whole time series in archive amplitude units. The *variogram score* compares temporal increments and is sensitive to dependence across time. We use exponent 0.5 and lags of 0.02, 0.1, 0.5, 1, 2 and 5 seconds, with equal weight across lags. A finite-ensemble variance correction is included.

The *time-frequency score* applies fair CRPS to log band RMS in moving time windows. It uses four-second Hann windows, a two-second step between windows, and frequency bands of 2–4, 4–8 and 8–12 Hz. It asks whether the model reproduces how band-specific amplitude evolves through the recording. These scores complement one another; their numerical scales are different and are not added together. Their definitions follow the proper-scoring framework discussed by Gneiting and Raftery (2007) and Scheuerer and Hamill (2015).

4.3 Filtering and uncertainty

We also assess sensitivity to a fixed filter applied to generated waveforms. The filter has passband edges at 2 and 12 Hz and uses a Butterworth design with order parameter four, applied forward and backward in second-order-section form. The observations are left unchanged. This comparison tests an output-processing choice; it does not assume that the filter reproduces the archive's original processing. Here, raw output means generated waveforms before this additional filter; it does not mean unprocessed instrument recordings. Raw output remains the

primary comparison.

Filtered results are available for all three runs of both neural models and for one empirical repetition. We report that difference in replication explicitly. A filter applied to one side of an evaluation changes the predictive distribution being assessed and can alter its ranking.

For each method, we average an event's score across its available runs, then average across events. Differences between methods are paired on the same events. We estimate pointwise 95% intervals by resampling the 99 validation proximity groups 2,000 times, retaining all records in each selected group. Event weights are equal within each resampled dataset. This calculation accounts for the defined groups and avoids counting repeated scores for one event as independent observations.

The intervals describe uncertainty conditional on the fitted models and generated ensembles. They do not include model-selection uncertainty, a population distribution of training runs, or corrections for multiple comparisons. They also cannot account for dependence that the proximity groups miss. Because validation informed model choices, the intervals should be read as descriptive comparisons rather than confirmatory tests.

5 Results

5.1 Observed and generated waveforms

Figure 1 shows the observed waveform and three conditional diffusion draws for each of three events. To select the events, we sort the 169 validation recordings by magnitude, breaking ties by origin time and event code, and take positions 18, 85 and 152. These positions lie near the 10th, 50th and 90th percentiles of the ordered sample. We use the first three draws from the first training run. Neither the events nor the draws are selected for their resemblance to the observations.

The examples contain a concentrated burst of motion followed by decay, with visible differences in peak amplitude and the length of the main burst. For the magnitude 3.1 event, all three displayed draws have a larger absolute peak than the observation. For the magnitude 2.1 event, the third draw is much smaller than the first two. These are conditional samples, not reconstructions of the observed event: the model receives event characteristics, but no part of its recorded waveform. The examples show the variation that the model produces; the scores below assess its distribution across all 169 events. Appendix F compares the three generation methods for the same events.

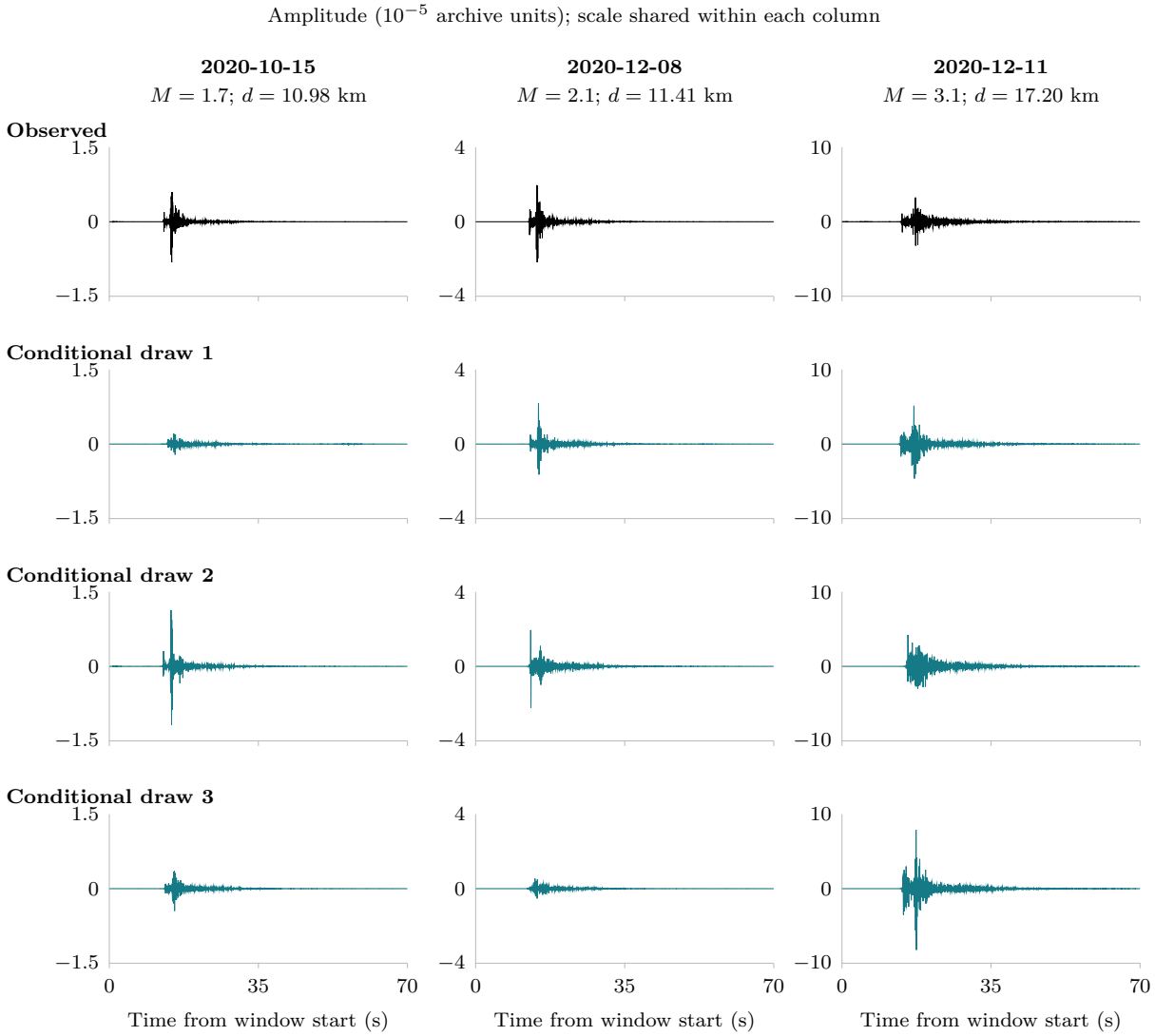


Figure 1: Observed waveforms and three conditional diffusion draws for events selected by magnitude rank. Black traces are observations; teal traces are the first three draws from run 1. All 3,500 samples are shown over the 70-second window. Amplitudes use archive units, with the multiplier shown above the panels. The vertical range is shared within each event column and differs across columns. No trace is individually normalized, shifted or additionally filtered. A generated trace need not match the phase or peak of the observation.

5.2 Conditional reference improves the matched diffusion model

Table 2 reports mean scores for raw output. Conditional reference lowers primary feature CRPS from 0.4933 to 0.4618, a reduction of 6.39%. The paired difference is −0.03152, with a 95% interval of $[-0.03831, -0.02486]$. Each of the three conditional runs has a lower feature score than its global-reference counterpart. Individual scores appear in Appendix C.

Table 2: Mean scores for raw generated waveforms. Lower is better in every column.

Model	Feature CRPS	Energy $\times 10^7$	Variogram $\times 10^7$	Time–frequency
Global reference	0.4933	7.5620	1.2737	0.3778
Conditional reference	0.4618	7.4803	1.1411	0.3361
Empirical retrieval	0.3778	8.1567	1.6292	0.4227

Each mean averages three repetitions on 169 events with 16 draws per event. Neural repetitions vary training and generation; empirical repetitions vary sampling. Energy and variogram scores retain their respective archive-unit scales.

The complementary scores generally point in the same direction (Table 3). Conditional reference has lower variogram and time-frequency scores, with pointwise intervals below zero. Its waveform energy score is also lower, but the interval's upper endpoint is very close to zero. The variogram and time-frequency results provide clearer evidence of an advantage than the energy score.

The feature breakdown in Figure 2 helps explain the aggregate improvement. Conditional reference improves mean timing, spectrum, envelope and autocorrelation scores. The amplitude difference is small and its interval includes zero. The improvement is therefore not simply an improvement in overall waveform amplitude.

Table 3: Paired score differences for raw output, with pointwise 95% intervals. Negative values favor conditional reference.

Score	Conditional − global	Conditional − empirical
Feature CRPS	$-0.03152\ [-0.03831, -0.02486]$	$0.08398\ [0.06133, 0.10214]$
Energy $\times 10^8$	$-0.8171\ [-2.0882, -0.0124]$	$-6.7635\ [-11.5434, -3.2632]$
Variogram $\times 10^8$	$-1.3263\ [-1.9439, -0.8229]$	$-4.8813\ [-8.5936, -2.2106]$
Time–frequency CRPS	$-0.04162\ [-0.05384, -0.03205]$	$-0.08659\ [-0.11968, -0.06279]$

Intervals resample whole proximity groups after averaging scores across repetitions. They condition on these fitted models and ensembles and do not adjust for selection or multiple comparisons.

5.3 Empirical retrieval changes the ranking across scores

Empirical retrieval obtains the lowest raw feature CRPS, 0.3778. Conditional diffusion exceeds it by 0.08398, with a 95% interval of $[0.06133, 0.10214]$. The empirical feature advantage appears in all three sampling repetitions.

This ordering reverses for the complementary waveform measures. Conditional diffusion has lower waveform energy, variogram and time-frequency scores than empirical retrieval, and all three pointwise difference intervals are below zero (Table 3). Empirical retrieval is therefore a strong feature-based benchmark, but it does not lead on every waveform measure.

Figure 2 shows where the feature ranking comes from. Empirical retrieval has a much lower spectral feature score: 0.3913, compared with 0.9716 for conditional diffusion. Conditional diffusion has lower amplitude, timing and envelope scores, while empirical retrieval has a slightly lower autocorrelation score. The spectral difference is large enough to outweigh the other groups

in the equally weighted average. This decomposition explains the aggregate ranking without changing the score or excluding unfavorable features.

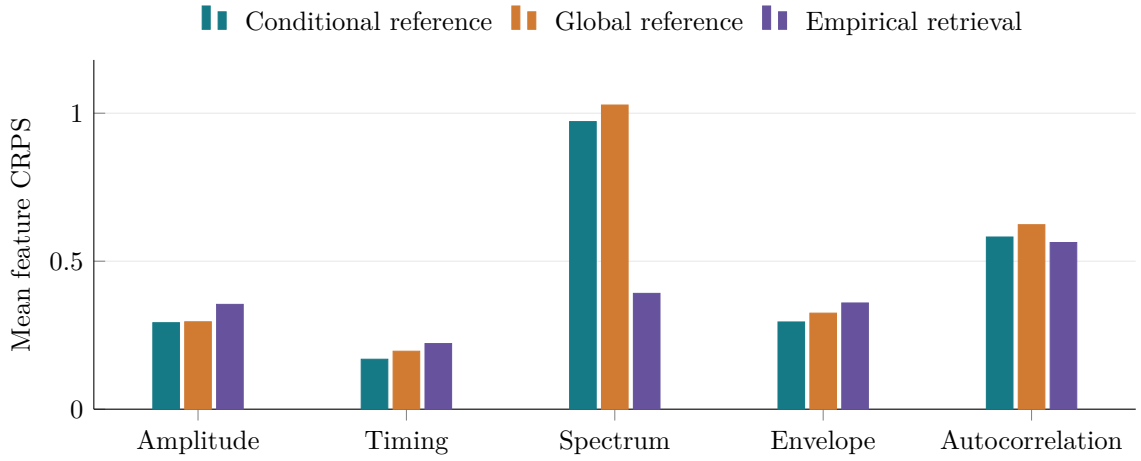


Figure 2: Raw feature scores by group. Each bar averages three repetitions on the same 169 events. The primary score gives equal weight to the five groups. Empirical retrieval has a large spectral advantage, while conditional diffusion has lower amplitude, timing and envelope scores. Bars show means, not uncertainty intervals.

The two spectral measures also answer different questions. The primary spectral features summarize frequency content over the full record and include frequencies outside 2–12 Hz. The time-frequency score follows amplitude through time within three bands inside that range. A favorable whole-record spectral distribution does not guarantee favorable local band amplitudes. Their different rankings reflect different aspects of the generated waveforms.

5.4 Filtering improves feature scores but not every waveform score

Figure 3 compares primary CRPS before and after filtering. The conditional mean falls from 0.4618 to 0.4344, and the global mean falls from 0.4933 to 0.4735. The conditional advantage remains in each run. Its mean reduction relative to global reference is 8.27%, with a paired difference of −0.03914 and interval [−0.04477, −0.03237].

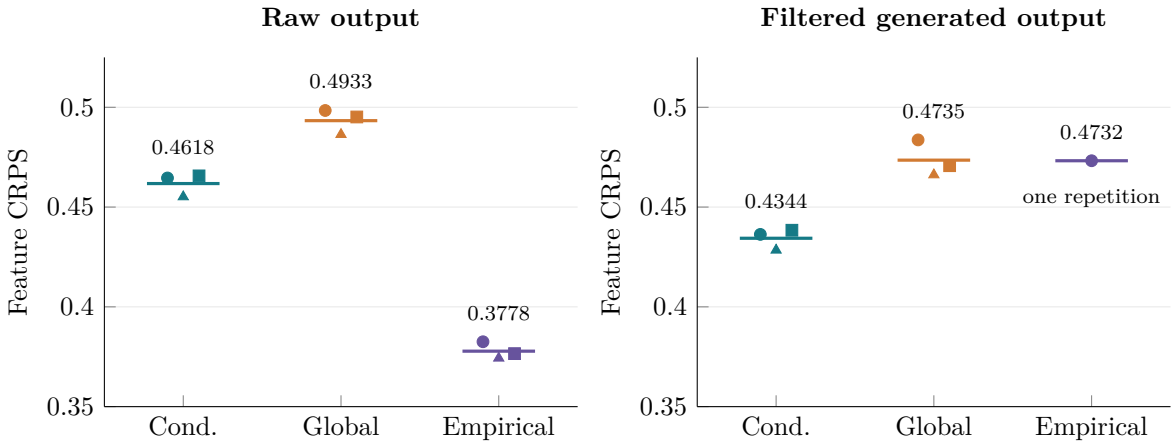


Figure 3: Primary feature CRPS before and after filtering generated waveforms. Markers show repetitions and horizontal lines their means; lower is better. All comparisons use 169 events and 16 draws per event. Filtered empirical retrieval has one repetition, while the other entries have three. The observed waveforms are unchanged.

The empirical filter comparison points in the opposite direction. For the one available repetition, its feature score rises from 0.3825 to 0.4732. The filtered conditional mean is lower by 0.03884, with an interval of $[-0.05629, -0.02385]$. This contrast averages three conditional runs against one empirical repetition. It does not measure the sampling variability of the filtered empirical model. The reversal also occurs when only the first conditional run is compared with the first empirical repetition.

Filtering is not a uniform improvement for the neural models. For conditional diffusion, it reduces primary CRPS by 0.02741 but increases waveform energy by 9.84×10^{-10} and time-frequency CRPS by 0.00210. The pointwise intervals for both increases lie above zero. Global reference shows the same directions. The changes in waveform energy are small relative to its score level, but their direction matters: a better feature score alone cannot establish that filtering improves the generated distribution in every relevant sense. Appendix D reports the paired filter effects.

After filtering, conditional reference continues to improve on global reference in variogram and time-frequency scores. The waveform energy difference is unresolved because its interval includes zero. This reinforces the need to distinguish the main feature result from the strength of evidence for each complementary measure.

6 Discussion

6.1 What the conditional reference contributes

The matched comparison supports an event-dependent reference for this archive. The same network performs better when amplitude, envelope and spectral predictions enter the coordinate transformation. The transformation may reduce predictable differences among events, leaving diffusion to learn the remaining variation. The comparison establishes a score difference, but it does not identify the mechanism or the contribution of each reference component. Separate component comparisons would be required for that purpose.

The reference provides a statistical description of waveform variation. The transformation is invertible and the diffusion model can represent non-Gaussian residual structure. This combination provides a tractable statistical reference while leaving the learned waveform distribution flexible.

6.2 What the score disagreement contributes

The empirical model is an important part of the comparison. Its strong feature score shows that resampling recorded waveforms can be difficult to beat when training and evaluation events share a restricted setting. At the same time, its worse waveform scores show that preserving recorded shapes does not ensure the best conditional distribution for a new event.

The scores evaluate different properties, so their disagreement is useful information. The intended use of the waveforms should guide which properties matter most. Whole-record features may matter for some tasks, whereas time-local amplitudes and temporal increments may matter for others. Reporting the measures together makes this tradeoff visible.

Filtering makes this point especially clear. It changes the generated distribution and can improve one score while worsening another. Reporting only the more favorable processing choice would conceal this sensitivity. Here, both choices are retained, with raw output as the main result and filtering as a separate sensitivity analysis.

6.3 Limits of the evidence

The most important limitation is validation reuse. The same set supported checkpoint selection and informed model development, so its scores may be optimistic. Repeating training does not create new evaluation events. Confirmation of the conditional-reference advantage requires recordings that have not influenced model choices.

The three neural runs provide limited information about stability. Software versions differ between the first pair and the remaining pairs, and training and sampling randomness vary together. The paired designs control these settings within each comparison, but do not separate their effects across runs. The uncertainty intervals also omit selection and training-run uncertainty and are not adjusted for the number of comparisons.

The scientific scope is local. All observations come from one station and a small-magnitude range. The results do not establish performance at other stations or for damaging earthquakes. Incomplete amplitude calibration and processing history also limit physical interpretation. The feature and variogram measures also capture only selected distributional properties, and 169 evaluation records limit the precision of all comparisons. The study does not establish calibrated tail probabilities or a universally best generator.

7 Conclusion

An event-dependent envelope and spectral reference improves waveform diffusion on the ARMT validation set. Across three runs, it reduces the primary feature CRPS by 6.39% relative to a global reference, with supporting differences in variogram and time-frequency scores. The advantage remains under a fixed filtering comparison.

The broader lesson concerns evaluation. Empirical retrieval performs better on the primary feature score, conditional diffusion performs better on the complementary waveform scores, and filtering changes the feature ranking. These results support conditional reference modeling while showing that conclusions about waveform quality depend on the properties being measured. Independent evaluation is needed to establish how well the observed advantage carries to new recordings.

Data availability

The ARMT waveform archive and event-level outputs are not publicly deposited. The paper reports the partition rules, model settings, score definitions and aggregate results. Reproduction of the waveform experiments requires access to the source archive.

A Model and sampling settings

The reference has 64 envelope bins and 64 spectrum bins. Ridge regression uses penalty 10, and the fitted conditional contribution is multiplied by 0.75. Standardized metadata are clipped to $[-8, 8]$ before forming linear and quadratic terms. Predicted envelope and spectral shape ratios are bounded by 32. Shapes are RMS-normalized, and the envelope includes the predicted amplitude. Log amplitude is restricted to the training target range extended by one unit at each end. The relative energy floor used to form reference targets is 10^{-8}.

The U-Net has four resolution levels with widths 64, 128, 256 and 256, two residual blocks per level, four-head attention at the coarsest level, and dropout 0.1. Residual convolutions have

kernel width five. Noise and metadata embeddings enter each residual block. Three input position channels encode normalized time and its sine and cosine. AdamW uses zero weight decay, gradient clipping at norm one, and a cosine learning-rate schedule. The exponential-average decay at step s is $\min\{0.999, (1+s)/(10+s)\}$.

EDM uses $\sigma_{\text{data}} = 0.5$, $\log \sigma \sim N(-1.2, 1.2^2)$ and weight

$$w(\sigma) = \frac{\sigma^2 + \sigma_{\text{data}}^2}{(\sigma \sigma_{\text{data}})^2}.$$

The denoiser uses the standard EDM skip, input and output scalings. Sampling starts from Gaussian noise at $\sigma_{\max} = 80$, follows a 25-level schedule with exponent seven down to $\sigma_{\min} = 0.002$, and ends at zero. Heun updates are used except for the final Euler update, giving 49 network evaluations per sample. These settings are shared across the two reference models.

B Feature definitions and secondary scores

The amplitude group contains log RMS, log absolute peak and log crest factor. Timing features are the times at which cumulative squared amplitude reaches 5%, 50% and 95%, together with the duration from 5% to 95%. Spectral features are log RMS Fourier magnitudes in 12 geometrically spaced bands from 0.5 to 22.5 Hz. The envelope group contains log RMS in 35 consecutive two-second blocks. The autocorrelation group uses the uncentered product average at lags of 0.02, 0.04, 0.1, 0.2, 0.5 and 1 second, divided by full-record mean squared amplitude. Amplitude, spectral and envelope quantities are floored at 10^{-12} before taking logs; the denominator of cumulative squared amplitude is floored at 10^{-24}. Feature standard deviations are floored at 10^{-3} before scaling.

For the variogram score, let $q_{k,t,h} = |x_{k,t+h} - x_{k,t}|^{1/2}$ and let $q^o_{t,h}$ be the corresponding observed increment. At each time and lag, the estimate is

$$(q^o_{t,h} - \bar{q}_{t,h})^2 - \frac{s^2_{t,h}}{K},$$

where s^2 is the unbiased variance across generated members. We average over valid time pairs and then over the six lags. The correction removes the finite-ensemble contribution to the squared error of the estimated increment moment. This score assesses increment moments and does not uniquely determine the full waveform law.

The time-frequency calculation uses a Hann periodogram with no detrending, integrates spectral density over the half-open bands $[2, 4)$, $[4, 8)$ and $[8, 12)$ Hz, and takes the natural log of band RMS. RMS is floored at 10^{-30}. Fair CRPS is averaged equally over time windows and bands. All secondary calculations restore the original common waveform amplitude scale before scoring.

The raw feature 5th–95th percentile coverage is 0.669 for conditional reference, 0.714 for global reference and 0.806 for empirical retrieval. Corresponding mean standardized widths are 1.208, 1.484 and 1.766. These are descriptive quantiles from 16 draws, not evidence of nominal 90% conditional coverage. They show that the conditional model's lower score does not coincide with broader predictive intervals.

C Scores by run

The first, second and third rows correspond to random seeds 0, 1 and 2. Each neural row changes training and generation randomness; each empirical row changes sampling from the

Table 4: Primary feature CRPS by repetition.

Run	Raw cond.	Raw global	Raw empirical	Filtered cond.	Filtered global
1	0.464572	0.498393	0.382515	0.436303	0.483651
2	0.455138	0.486355	0.374305	0.428417	0.466144
3	0.465626	0.495140	0.376567	0.438385	0.470730
Mean	0.461779	0.493296	0.377796	0.434368	0.473508

Filtered empirical results are available for the first repetition only, with score 0.473211. The raw and filtered neural scores in each row use the same generated draws before and after filtering.

same training bank. For raw conditional minus global feature CRPS, the three paired intervals are $[-0.04163, -0.02473]$, $[-0.03901, -0.02314]$ and $[-0.03833, -0.02229]$. After filtering, they are $[-0.05482, -0.03722]$, $[-0.04475, -0.03029]$ and $[-0.04006, -0.02420]$. These intervals use the same group resampling scheme as the main results.

D Effects of filtering

Table 5: Filtering effects: filtered minus raw score, with pointwise 95% intervals. Negative values indicate an improvement after filtering.

Score	Conditional reference	Global reference
Feature CRPS	-0.02741 $[-0.04260, -0.01100]$	-0.01979 $[-0.03278, -0.00467]$
Energy $\times 10^8$	0.0984 $[0.0097, 0.1833]$	0.0885 $[0.0102, 0.1882]$
Variogram $\times 10^8$	0.0273 $[-0.0177, 0.0880]$	0.0543 $[0.0063, 0.1273]$
Time–frequency CRPS	0.00210 $[0.00018, 0.00444]$	0.00259 $[0.00106, 0.00450]$

Both comparisons average the paired changes across three runs. Observed recordings are unchanged. Differences on different score scales should not be compared by their absolute size.

E A comparison with latent-spectrogram diffusion

A supplementary comparison uses an adaptation of the principal HighFEM latent-spectrogram model (Palgunadi et al., 2025). It uses the same 774 training records, 169 validation records, seven event inputs and 50 Hz vertical component as the main analysis. The original model's three-component setting is reduced to one component. A variational autoencoder represents spectrograms, a latent diffusion model generates their codes, and Griffin–Lim phase reconstruction returns them to waveforms.

The autoencoder and diffusion stages each train for 300 epochs. Generation uses 25 EDM sampling steps and 128 Griffin–Lim iterations. We compare its single fitted model with the first conditional-reference run, using 16 draws per event and the same score definitions. The architectures and training budgets differ, so this is a model comparison rather than a controlled reference ablation.

The raw HighFEM-minus-conditional difference is 0.01035, with interval $[-0.00025, 0.02496]$. After filtering, it is 0.01149, with interval $[0.00128, 0.02186]$. The raw feature comparison is therefore unresolved, whereas the filtered feature comparison favors conditional reference. All three complementary waveform difference intervals include zero under both output choices. This single-run comparison does not establish a general advantage over HighFEM, but it provides additional evidence that the conclusion depends on the score and output processing.

Table 6: Single-run feature CRPS for the HighFEM adaptation and conditional reference. Lower is better.

Model	Raw output	Filtered output
HighFEM adaptation	0.474925	0.447798
Conditional reference, run 1	0.464572	0.436303

F Waveform examples across models

Figure 4 shows the first draw from each model for the same events as Figure 1, using the same amplitude ranges. Empirical retrieval returns a complete training recording; diffusion generates new traces. These examples illustrate differences in shape and amplitude. A single draw cannot establish which conditional distribution is better.

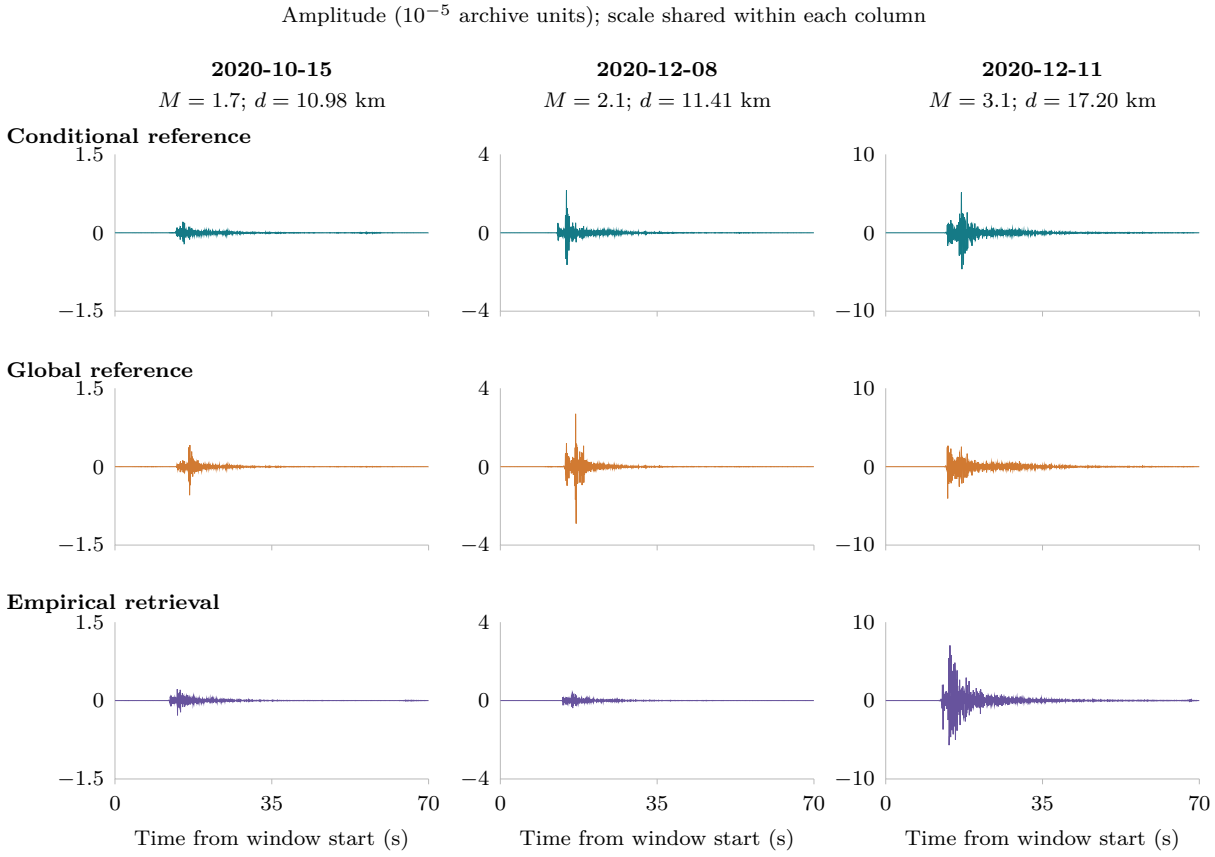


Figure 4: Waveform examples across generation methods. Each column shows the first conditional, global and empirical draw from repetition 1 for one event. The corresponding observations appear in Figure 1. All 3,500 samples are retained, with a shared vertical range within each column and amplitudes in archive units. No trace is individually normalized, shifted or additionally filtered. Empirical traces are sampled from the training archive.

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